



Accessibility Self-Study Tutorial



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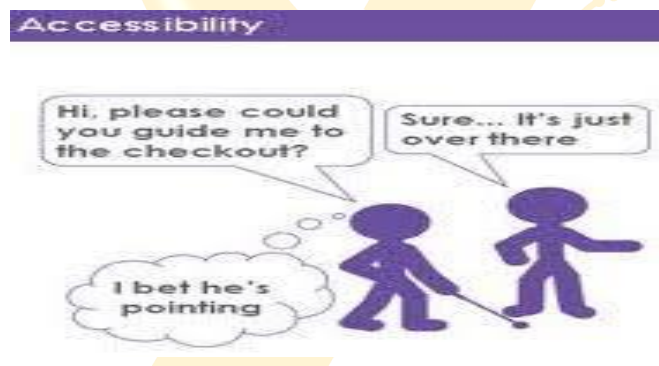
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What is accessibility?

Accessibility is ability to access or reach anything.



Talking on technical front, Accessibility involves two key issues: how users with disabilities access electronic information and how web content designers and developers enable web pages to function with assistive devices used by individuals with disabilities.

For the user with a disability, the challenge is to identify tools that provide the most convenient access to web-based and other electronic information. For the web content designer/developer, the challenge is to remove the obstacles that prevent accessibility tools from functioning effectively. In many cases, these challenges are relatively simple to overcome, but sometimes the solutions require some additional thought and effort.

Accessibility Testing



Accessibility Testing is a subset of Usability Testing, and it is performed to ensure that the application being tested is usable by people with disabilities like hearing, color blindness, old age and other disadvantaged groups.

Defining disabilities

The range of disabilities is broad and difficult to categorize, however here we will provide you some sense of different type of disabilities:



Vision impairment:

Visual impairments covers a wide spectrum of conditions – from ‘low-vision’ or ‘partially sighted’ through to ‘total blindness’. There are other visual conditions such as the different forms of color blindness.

- **Low vision or partial sight:**

People with low vision may suffer from extreme forms of short-sightedness or long-sightedness, or their vision may be blurred or unclear in other ways.

When interacting with websites people with low vision may typically need to do one or more of the following:

- Adjust the size of the text on their browser
- Use assistive software to magnify the web page
- Override the style of web pages to use high contrast color schemes
- Use screen readers and/or braille display

The following features of web pages can make life difficult for people with low vision:

- Fixed height text that cannot be resized
- Semantically connected items that are some distance apart – e.g. form labels and input boxes too far apart
- Insufficient color contrast
- Busy page backgrounds
- Use of color only to convey meaning
- Dynamic updating of content some distance from focus

- **Blindness:**

Blindness itself has degrees and there is a legal measure in a number of countries. People who are 'legally blind' may still have some residual vision or only the ability to tell light from dark. Typically blind users will have extreme difficulty using websites as sighted people would.

When interacting with websites people with blindness may typically need to do one or more of the following:

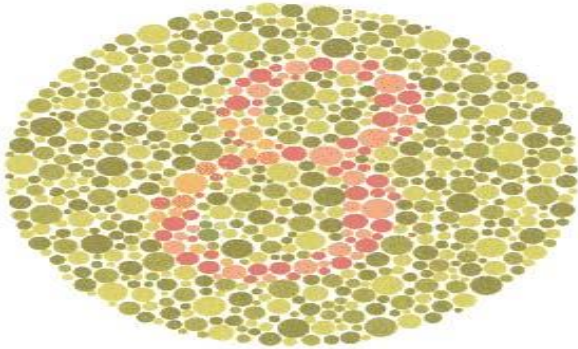
- Use a screen reader
- Use a braille display

The following features of web pages can make life difficult for people with blindness:

- Images without appropriate alternate text
- Badly or inappropriately marked up pages, forms etc.
- Poorly worded links
- Pages with incorrect or non-unique titles
- Pages without headings
- PDFs that are not created with accessibility in mind
- AJAX updates of pages
- Flash, Java and other RIA functionality

Accessible Minds

- **Color blindness:**



A common misconception is that people with color blindness only see in black and white (or greyscale). Whilst that is true of one form there are many types of color blindness and each one has differing effects on people's perception of color. Color blindness is much more common in men than women.

Mostly when interacting with websites people with color blindness won't need to use any special techniques or software. However the following features of web pages can make life difficult for people with color blindness:

- Insufficient color contrast between text and background
- The use of colors alone to indicate meaning

Deaf or hard of hearing:

Hearing impairments can range from mild to profound. People who are hard of hearing may use a range of strategies and equipment including speech, lip-reading, writing notes, hearing aids or sign language interpreters.

The following features of web pages can make life difficult for people with deafness (not an exhaustive list):

- Video content with no captioning
- Audio content or signals with no captioning
- Background noises too loud
- Very complex and/or obscure language

Mental health conditions:

Mental illness is a general term for a group of illnesses that affect the mind or brain. These illnesses, which include bipolar disorder, depression, schizophrenia, anxiety and personality disorders, affect the way a person thinks, feels and acts.

The following features of web pages can make life difficult for people with cognitive impairments

- Long sentences and complex language
- Insufficient instructions on how to use functionality (forms, applications etc) or what they're for
- Pages that require quick response
- Sites that have inconsistent page layout/navigation
- Sites that change layout or structure frequently
- Pages that update content unexpectedly

Motor disability:

Motor impairments take many forms and I don't intend to cover all of them here. But these impairments can affect the use of people upper limbs and hands – the way most people interact with a computer and web.

People with motor impairments may need to do one or more of the following to access the web:

- Rely solely on keyboard interaction with web pages
- Use specialized keyboards, switches or other input devices
- Use voice recognition software
- Use head pointers or eye gaze systems

Accessible Minds

Assistive technologies

Users with disabilities frequently rely on hardware and software to access web content. These tools, known as assistive technologies, range from screen readers to touchscreens and head pointers.



- **Visually impaired user** of the web frequently use software called a screen reader to read the contents of a web page out loud. Two common screen readers are **JAWS(Licensed tool)** from Freedom Scientific and **NVDA(open source)**. Screen readers enable users to hear, rather than read, the contents of a web page; however, a screen reader can read only text, not images or animations. It is desirable, therefore, that images and animations be assigned text descriptions that screen readers can read. These text descriptions are called alternative text, or alt text.
- **Users with low vision** often depend on magnification software. Some magnifier users utilize speech features within magnification tools or use magnifier software in conjunction with a screen reader. Two popular screen magnifiers are **ZoomText** from Ai Squared and **MAGic** from Freedom Scientific. Screen magnifiers increase the size of a portion of the user's screen or the entire screen in order to make the content more visible. Screen magnifiers that provide a speech feature depend on the same accessibility information as screen readers.
- **Users with mobility issues** may rely on the keyboard instead of the mouse to navigate web pages. For individuals with nerve damage, arthritis, or repetitive motion injuries, use of the mouse may not be comfortable or possible. **Using only Tab and Enter on the keyboard**, it is possible for these individuals to navigate a page with ease. Many users of the Internet have the capability to navigate without a mouse and are simply unaware of it.

In some cases, users may employ touchscreens, head pointers, or other assistive devices. A touchscreen allows an individual to navigate the page using her or his hands without the fine-motor control required by the mouse. A head pointer is simply a stick placed in a person's mouth or mounted on a head strap that the person uses to interact with a keyboard or a touchscreen.

In these cases, it is very important that essential components of the page work without a mouse. Rollovers, pull-down lists, and interactive simulations are all examples of elements that typically depend on the mouse for user interaction. The designer or developer of these elements must ensure that keyboard-defined events are included along with mouse-defined events.

Types of Accessibility Testing

Screen Reader Testing:

Many people with visual impairments use keyboard input and screen readers to consume web content, relying on technology that converts text into synthesized speech. In effect, the computer “reads” the text out loud for the user.



Recommended Screen Reader and Browser Combinations

Different screen readers tend to be optimized to work with different browsers. Though it is possible that this information may change, at this time, the following combinations work best together:

- Windows: Chrome with NVDA/JAWS as per request
- Mac: Safari and VoiceOver
- iPhone: Safari and VoiceOver
- Android: Chrome with TalkBack

Keyboard Only Testing:



Testing with a keyboard is an essential part of any accessibility evaluation. The basics of keyboard testing are simple—the Tab key can be used to navigate through links and form controls, Enter (and sometimes Spacebar) selects an element, and the arrow keys are sometimes used for other navigation. However, when testing with a keyboard, you are not just trying to interact with the page successfully, you also want to ensure all interactions are predictable. This requires an understanding of common keyboard interactions.

Color Contrast Ratio Validation:



Level AA requires a contrast ratio of at least 4.5:1 for normal text and 3:1 for large text. Level AAA requires a contrast ratio of at least 7:1 for normal text and 4.5:1 for large text.

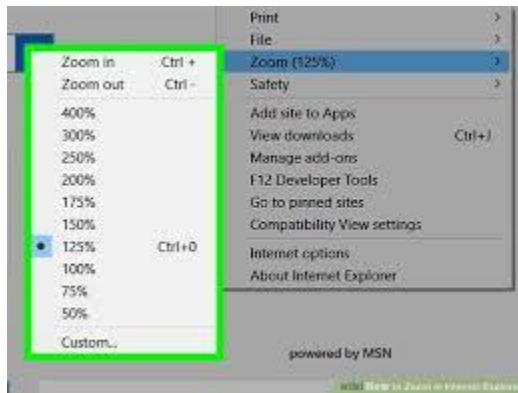
Large text is defined as 14 point (typically 18.66px) and bold or larger, or 18 point (typically 24px) or larger.

The Color Contrast Check Tool allows to specify a foreground and a background color and determine if they provide enough of a contrast "when viewed by someone having color deficits or when viewed on a black and white screen".

The tool will indicate that the colors pass the test if both the color difference and the brightness difference exceed their threshold. It will indicate that it sort of passes if only one of the two values exceed their threshold. And finally, it'll fail to pass if neither value exceeds its threshold.

For non-text contrast (elements like button, combo box, radio buttons, checkbox etc.), required contrast ratio is at least 3:1

Browser Zoom/Text Zoom Testing:



Low vision user increases the text size on a Web page in a browser up to 200 percent in order to read text at small size by zooming it to larger extent.

Except for captions and images of text, text can be resized without assistive technology up to 200 percent without loss of content or functionality. All the content scales uniformly, and if necessary, browser provides scroll bars.

High Contrast Testing:

High contrast settings benefit users with low vision or other visual disabilities. The "High Contrast Themes" present in Windows change or invert web page background and text colors. To enable High Contrast in Windows 7, select the Start menu (Windows Key), enter "Contrast" in the search field, and select "Turn High Contrast on or off".

High contrast testing is highly recommended for testing web content in Windows High Contrast Mode. There is no special software to purchase (assuming you have access to the Windows operating system) and almost no learning curve, and It helps uncover accessibility issues that cannot easily be uncovered in another way. Testing with Windows High Contrast is especially important if you are using non-decorative CSS images.



High Contrast Items to Verify:

1. Verify all UI elements and controls are visible
If the background changes to black and the font color is hard-coded black, the application or dialog is not usable.

2. Verify differentiable appearance of inactive and active selection
3. Verify visual focus is available in high contrast mode
Focus must be shown at all times. A great way to find bugs is to tab around the application. If focus ever disappears, it is an Accessibility bug.
4. Verify multiple schemes function
All color schemes should work, since font sizes and colors come from the OS. However, due to time and resource constraints, it is important to test with the most commonly-used schemes, like High Contrast Black (large). With some certainty, if the application responds correctly to one scheme, it in theory responds to all, because it is reading the color data from the same location. Only the data is changing values.
5. Verify no text truncation occurs

Caption, Transcript and Audio Description validation:

Captions:

Captions are text versions of the spoken word presented within multimedia. Captions allow the content of web audio and video to be accessible to those who do not have access to audio. Though captioning is primarily intended for those who cannot hear the audio, it has also been found to help those that can hear audio content, those who may not be fluent in the language in which the audio is presented, those for whom the language spoken is not their primary language, etc.

Common web accessibility guidelines indicate that captions should be:

- **Synchronized** - the text content should appear at approximately the same time that audio would be available
- **Equivalent** - content provided in captions should be equivalent to that of the spoken word
- **Accessible** - caption content should be readily accessible and available to those who need it

Transcripts:

Transcripts also provide an important part of making web multimedia content accessible. Transcripts allow anyone that cannot access content from web audio or video to read a text transcript instead. Transcripts do not have to be verbatim accounts of the spoken word in a video. They should contain additional descriptions, explanations, or comments that may be beneficial, such as indications of laughter or an explosion. Transcripts allow deaf/blind users to get content through the use of refreshable Braille and other devices. For most web video, both captions and a text transcript should be provided. For content that is audio only, a transcript will usually suffice.

Audio Descriptions:

Audio descriptions are intended for users with visual disabilities. They provide additional information about what is visible on the screen. This allows video content to be accessible to those with visual disabilities. Though not commonly utilized in television and movies, it is gaining in popularity. Audio descriptions are helpful on the web if visual content in web video provides important content not available through the audio alone.

Automated Testing Tools



Although using an exploratory-type testing approach conducted by folks that actually have impairments will give you some of the best coverage, but partial automation of accessibility testing is also possible. Automated tool can only capture around 30% accessibility issues. Tool majorly provides warning and suggestions for best code writing practice.

There are N number of automated accessibility tools available in market like WAVE, AXE, A11Y Machine etc.

Why is accessibility important?

Accessibility is an immensely important effort. Individuals and organizations with different functions will find different reasons to adopt accessibility. The following is a list of reasons many find compelling.

- **Accessibility is the right thing to do**

Accessibility represents an important step toward independence for individuals with disabilities. Accessible web pages provide access to fundamental government services and information such as tax forms, social programs, and legislative representatives. Accessible web pages also make possible a broader range of employment and educational opportunities by providing added means of communication. In addition, accessibility allows users with disabilities to participate in day-to-day activities many of us take for granted, such as reading a newspaper or buying a gift for a loved one.

- **Accessibility is the law for many institutions**

With new national requirements in the United States, Canada, and the European Union, and more to come in the near future, there are numerous legal mandates for accessibility. These policies will likely expand in scope. In the United States, for instance, Section 508 of the U.S. Rehabilitation Act sets standards for web pages designed or maintained by federal agencies.

- **Accessibility offers benefits for all users**

As with many improvements intended for individuals with disabilities, the enhancements of accessible design offer benefits for all users of the web. Anyone who has pushed a shopping cart out of a grocery store can attest to the value of automatic doors and ramps cut into curbs.

Accessibility standards

Accessibility standards help designers and developers of web content identify and address accessibility issues.

[Section 508](#), is an amendment to the United States Workforce Rehabilitation Act of 1973, is a federal law mandating that all electronic and information technology developed, procured, maintained, or used by the federal government be accessible to people with disabilities.

[The Web Content Accessibility Guidelines \(WCAG\) 2.1](#) developed through the World Wide Web Consortium's (W3C's) Web Accessibility Initiative (WAI), are a series of guidelines for improving web accessibility to make websites and content accessible to users with disabilities and different devices (e.g. mobile phones). The previous version, WCAG 2.0, was published in December 2008 and became an ISO standard in October 2012.

WCAG standards consist of three conformance level: Level A, Level AA, and Level AAA.



Difference between A, AA & AAA:

- **Level A** (minimum) – the most basic web accessibility features. The website **MUST** satisfy this checkpoint. Otherwise, one or more groups will find it impossible to access information in the document. Satisfying this checkpoint is a basic requirement for some groups to be able to use web documents.
- **Level AA** (mid-range) – deals with the biggest and most common barriers for disabled users. Website **SHOULD** satisfy this checkpoint. Otherwise, one or more groups will find it difficult to access information in the document. Satisfying this checkpoint will remove significant barriers to accessing Web documents.
- **Level AAA** (highest) – the highest level of web accessibility. A website **MAY** address this checkpoint. Otherwise, one or more groups will find it somewhat difficult to access information in the document. Satisfying this checkpoint will improve access to web documents.

The [Americans with Disabilities Act \(ADA\)](#), is a civil rights law that “prohibits discrimination against individuals with disabilities in all areas of public life, including jobs, schools, transportation, and all public and private places that are open to the general public” (ADA National Network). The law guarantees equal opportunity for individuals with disabilities in public accommodations, employment, transportation, state and local government services, and telecommunications.

What is Difference between 508, ADA and WCAG 2.1?

- [Section 508](#) requires that accommodations be made for federal employees and members of the public with disabilities who wish to access **government Information and Communications Technology** (ICT), including government websites and multimedia.
- [ADA](#) requires that **businesses, state and local governments, and nonprofit services providers** make accommodations for the **disabled public** to access the same services as able-bodied patrons. Movie theaters, digital media, websites, and ICT are required to be accessible to people with disabilities.
- [WCAG](#) is simply a set of formal guidelines on how to develop accessible web content. These standards represent a higher level of accessibility than 508 standards, although they are identical or very similar. **WCAG focuses primarily on HTML accessibility.**